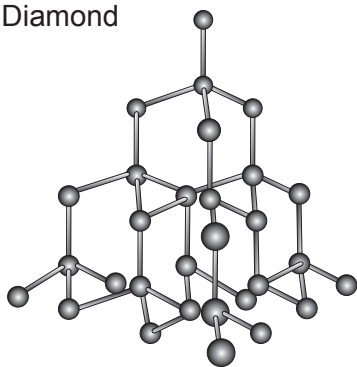


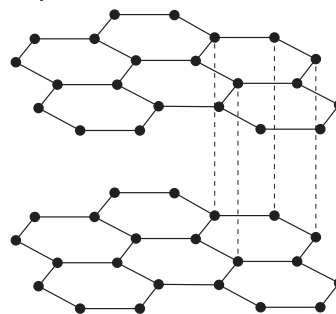
6. Carbon exists as a number of different allotropes. These allotropes have some important commercial uses. The uses of these allotropes depends upon the structure.

(a) Two allotropes of carbon are diamond and graphite.

Diamond



Graphite



Use the diagrams and your knowledge to answer the following questions.

- (i) **Describe** the bonding between a carbon atom and its nearest neighbours in diamond.  
**Include** a dot and cross diagram of one carbon atom and its nearest neighbours in your answer. [3]

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- (ii) State the difference in the bonding between carbon atoms in diamond and graphite. [1]

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- (iii) Explain why graphite is a good conductor of electricity but diamond is a good insulator of electricity. [3]

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- (iv) Explain, in terms of structure and bonding, why graphite is soft and slippery but diamond is hard. [3]

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- (b) Graphene is a single layer of graphite. It can be rolled into the shape of a cylinder which is called a carbon nanotube.

The table gives some data about carbon nanotubes, steel and aluminium.

| Material        | Strength (MPa) | Density (g/cm <sup>3</sup> ) | Maximum length of power line before it breaks (km) | Electrical conductivity ( $\times 10^6$ units) |
|-----------------|----------------|------------------------------|----------------------------------------------------|------------------------------------------------|
| carbon nanotube | 62 000         | 1.3                          | 4700                                               | $\gg 1000$                                     |
| steel           | 500            | 8.0                          | 6.4                                                | 7                                              |
| aluminium       | 80             | 2.7                          | 1.0                                                | 40                                             |

Electricity power lines are currently made from a combination of steel and aluminium. Explain why carbon nanotubes would be better. [3]

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